COL703/
7203

Natural Deduction

30 July

$$\frac{\phi_1 \quad \phi_2}{\phi_1 \wedge \phi_2} \wedge i$$

$$\frac{\phi_1 \wedge \phi_2}{\phi_1} \wedge e_1 \qquad \frac{\phi_1 \wedge \phi_2}{\phi_2} \wedge e_2$$

$$\frac{\phi}{\neg\neg\phi} \qquad \frac{\neg\neg\phi}{\phi}$$

$$\frac{\boxed{\begin{array}{l} \phi \text{ assumption} \\ \vdots \\ \psi \end{array}}}{\phi \to \psi} \to i$$

$$\frac{\phi \quad \phi \to \psi}{\psi} \text{ MP}$$

$$\frac{\phi \to \psi \quad \neg\psi}{\neg\phi} \text{ MT}$$

$\neg q \to \neg p \quad \vdash \quad p \to \neg\neg q$

$\phi \qquad \psi$

$\neg q \to \neg p$ premise

p assumption

$\neg\neg p$ $\neg\neg i$

$\neg\psi$

$\neg\neg q$

$p \to \neg\neg q$

$\vdash p \to p$

p assm

$p \to p$

$p \land q \to r \vdash p \to (q \to r)$

0. $p \land q \to r$ premise
1. p assumption
2. q assumption
3. $p \land q$ $\land i$ 1,2
4. r MP 3,0

$q \to r$

$p \to (q \to r)$

$$p \to (q \to r) \dashv\vdash p \land q \to r$$

0.	$p \to (q \to r)$	
1.	$p \land q$	assumption
2.	p	$\land e_1$, 1
3.	$q \to r$	MP 2, 0
4.	q	$\land e_2$, 1
5.	r	MP, 4, 3

$p \land q \to r$

$p \to q \vdash p \wedge r \to q \wedge r$

$p \wedge r$ assume

p $\wedge e$

$\vdots$

Disjunction

$$\frac{\phi}{\phi \vee \psi}\ \vee i_1 \qquad\qquad \frac{\psi}{\phi \vee \psi}\ \vee i_2$$

$$\frac{\phi \vee \psi \qquad \boxed{\begin{array}{c}\phi \\ \\ \\ \chi\end{array}} \quad \boxed{\begin{array}{c}\psi \\ \\ \\ \chi\end{array}}}{\chi}\ \vee e$$

$$p \vee q \vdash q \vee p$$

$$\frac{p \vee q \qquad \boxed{\begin{array}{c} p \\ \\ q \vee p \end{array}} \qquad \boxed{\begin{array}{c} q \\ \\ q \vee p \end{array}}}{q \vee p}$$

$$p \vee q$$

$q \to r \vdash p \vee q \to p \vee r$

$p \vee q$ assumption

p assume

$p \vee r$

q assume

r MP

$p \vee r$

$p \vee r$

$p \vee q \to p \vee r$

$(p \lor q) \lor r \vdash p \lor (q \lor r)$

$(p \lor q) \lor r$

premise

$p \lor q$

p $\quad$ q

r

$q \lor r$

$p \lor (q \lor r)$

$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$

$p \wedge (q \vee r)$ premise

$- \; p$

$- \; q \vee r$

$\boxed{\begin{matrix} q \\ p \wedge q \end{matrix}}$ $\boxed{\begin{matrix} r \\ p \wedge r \end{matrix}}$

Exercise

$$p \vee (q \wedge r) \vdash (p \vee q) \wedge (p \vee r)$$

What about negations?

Contradictions

$$\left.\begin{array}{l} \phi \wedge \neg\phi \\ \neg\phi \wedge \phi \end{array}\right\}$$

denoted as $\perp$ (bottom)

Contradictions allow us to derive anything.

$$\frac{\perp}{\phi}\ \perp e \qquad\qquad \frac{\phi \quad \neg\phi}{\perp}\ \neg e$$

$$p \wedge \neg p \vdash q$$

Exercise

$$\neg p \vee q \vdash p \to q$$

p ass

$\neg p \vee q$

$\neg p$

$\bot \quad \neg e$

$q \quad \bot e$

q

q copy

q

$p \to q$

Introduce negations

$$\dfrac{\boxed{\begin{array}{c}\phi \\ \\ \\ \bot\end{array}}}{\neg\phi}\ \neg i$$

$$\boxed{\begin{array}{c}\phi \\ \boxed{\begin{array}{c}\psi \\ \\ \\ \bot\end{array}} \\ \neg\psi\end{array}}$$

Exercise

i) $p \to q,\ p \to \neg q \vdash \neg p$

ii) $p \to \neg p \vdash \neg p$

iii) $p \wedge \neg q \to r,\ \neg r,\ p \vdash q$